Model	verbose	placement	default	terse	memory.md
Claude Opus 4.6	0.6425	0.4670	0.4330	0.0540	0.0445
Claude Sonnet 4.6	0.2560	0.1055	0.1130	0.0125	0.0340
Gemini 3.1 Pro	0.6825	0.7115	0.7030	0.6945	0.6560
Gemini 3.1 Flash-Lite	0.2955	0.1420	0.2305	0.2405	0.0710
GPT-5.4	0.6175	0.4905	0.4200	0.1695	0.0830
GPT-5.4 nano	0.0575	0.0580	0.0510	0.0460	0.0395

Table 5 Per-digit accuracy under five confidentiality instructions (chance 0.1000), N=2, n=1000 per cell, 10 held-out secrets, single seed. The decoder is trained on `default` only and is never retrained. Columns are ordered by how forcefully the instruction protects the secret. Grok 4 and Grok 4.1 Fast are omitted: xAI retired those model identifiers on 2026-05-15 and they silently redirect, so runs generated afterwards cannot be attributed to the named model.

Model	verbose	placement	default	terse	memory.md
Claude Opus 4.6	990	993	1000	955	891
Claude Sonnet 4.6	1000	984	998	998	994
Gemini 3.1 Pro	1000	1000	1000	1000	1000
Gemini 3.1 Flash-Lite	998	1000	993	998	996
GPT-5.4	904	824	846	787	725
GPT-5.4 nano	681	850	918	798	847

Table 6 Records used for the suppression measurement (out of 1000). A record is dropped iff its response contains no parseable N-digit number.

H Confidentiality-Instruction Mismatch

The adversary in Section 3.1 estimates response distributions using one fixed confidentiality instruction. A deployment may not use that instruction. To test whether this matters, we take each model's decoder trained *only* on the default instruction and score it, with no retraining, on transcripts generated under four others: a terse one-liner; a long formal policy that forbids output, paraphrase, encoding or transformation; a variant placing the secret after the instructions; and a `USER.md`-style memory file. All five are reproduced verbatim in Appendix D.

First, **the decoder transfers without adaptation**: Gemini 3.1 Pro reads the secret at 0.703 per-digit under the default instruction and never drops below 0.656 under any of the five, against a chance rate of 0.1. Second, **leakage tracks instruction emphasis**: four of the five leaky models are emphasis-dependent, with Claude Opus 4.6 spanning 0.643 under the verbose policy down to 0.054 under the terse one-liner, and GPT-5.4 spanning 0.617 to 0.083. Only Gemini 3.1 Pro is instruction-invariant. Contrary to what one might expect of an aligned model, per-digit accuracy *rises* with how forcefully the instruction protects the secret, and falls only when the instruction is too casual to induce suppression at all. Third, **the resistant tier is anti-informative**: GPT-5.4 nano reads 0.040–0.058 per digit against a chance rate of 0.1, and its channel entropy of 10.65–11.52 bits sits *above* the 6.64-bit no-leakage baseline, meaning the trained decoder does worse than a uniform guess.

We remark that exact match at N=2 spans only the ten held-out secrets and is reported for completeness; per-digit accuracy and channel entropy are the main metrics. The suppression column has a different denominator from the accuracy columns since a record is dropped when the response contains no parseable N-digit number (Table 6); the two should not be read as sharing an n.

I Co-Present Distractors

The numeric setting of Section 3.1 places one secret in an otherwise minimal context. A real deployment holds many values, only some of them sensitive, which raises the question of whether the decoder recovers *the protected value* or merely *some number in the context*. We vary two factors independently: **salience**, whether the secret is introduced by an explicit "secret code" assignment or as one line item in a `USER.md` memory file, and **clutter**, whether the context holds three same-length distractors or six mixed-length numbers embedded in a prose biography. Cells A–D are the four combinations.

Model	A: explicit, minimal secret	A: explicit, minimal distr.	B: explicit, clutter secret	B: explicit, clutter distr.	C: `USER.md`, minimal secret	C: `USER.md`, minimal distr.	D: `USER.md`, clutter secret	D: `USER.md`, clutter distr.
Claude Opus 4.6	0.379 ± 0.253	0.0073	0.347 ± 0.259	0.0075	0.233 ± 0.116	0.0074	0.029 ± 0.036	0.0146
Claude Sonnet 4.6	0.014 ± 0.018	0.0100	0.015 ± 0.012	0.0113	0.049 ± 0.012	0.0119	0.005 ± 0.006	0.0092
Gemini 3.1 Pro	0.539 ± 0.351	0.0018	0.542 ± 0.218	0.0013	0.618 ± 0.250	0.0029	0.604 ± 0.347	0.0042
Gemini 3.1 Flash-Lite	0.258 ± 0.097	0.0070	0.290 ± 0.101	0.0112	0.139 ± 0.019	0.0048	0.003 ± 0.008	0.0110
GPT-5.4	0.488 ± 0.040	0.0150	0.446 ± 0.086	0.0172	0.319 ± 0.074	0.0119	0.011 ± 0.025	0.0099
GPT-5.4 nano	0.000 ± 0.000	0.0074	0.000 ± 0.000	0.0124	0.000 ± 0.000	0.0106	0.001 ± 0.003	0.0108

Table 7 Protected-secret exact match (mean $\pm$ sd over 5 seeds) against mean co-present distractor exact match, N=2, chance 0.01. Seeds rotate which secrets are held out. Distractors remain at chance in every cell. Grok 4 and Grok 4.1 Fast are omitted due to model retirement.

Model	A: explicit, minimal secret	A: explicit, minimal distr.	B: explicit, clutter secret	B: explicit, clutter distr.	C: `USER.md`, minimal secret	C: `USER.md`, minimal distr.	D: `USER.md`, clutter secret	D: `USER.md`, clutter distr.
Claude Opus 4.6	0.014 ± 0.003	0.0002	0.022 ± 0.014	0.0001	0.003 ± 0.002	0.0001	0.003 ± 0.000	0.0002
Gemini 3.1 Pro	0.023 ± 0.004	0.0001	0.016 ± 0.004	0.0001	0.016 ± 0.004	0.0001	0.012 ± 0.001	0.0002

Table 8 Protected-secret exact match (mean $\pm$ sd over 3 seeds) against mean co-present distractor exact match, N=4, chance 0.0001. Seeds rotate which secrets are held out. Distractors remain at chance in every cell. The N=4 arm is run at n=50,000 ($\sim$993 held-out secrets), which escapes the 10-held-out-secret ceiling that makes N=2 exact match noisy. Grok 4 and Grok 4.1 Fast are omitted: xAI retired those model identifiers on 2026-05-15 and they silently redirect, so runs generated afterwards cannot be attributed to the named model.

The attack is sharply secret-specific. Co-present distractors sit at chance in every cell of both designs (0.0013–0.0172 against a chance rate of 0.010 at N=2), while the protected value is recovered far above chance in the four leaky models. The design is uninformative for Claude Sonnet 4.6 and GPT-5.4 nano, which have no channel even in the base cell.

Salience and clutter interact in a model-independent manner. Gemini 3.1 Pro is undiminished across all four cells (0.539/0.542/0.618/0.604), so for that model the effect survives the transition from an explicit "secret code" assignment to a buried `USER.md` line entirely. Claude Opus 4.6 degrades gradually (0.379 to 0.029), while for GPT-5.4 and Gemini 3.1 Flash-Lite the two stressors together close the channel (0.488 to 0.011 and 0.258 to 0.003). Neither factor alone does so. Within the leaky models, the emission rates show the mechanism. The protected value is emitted far less often than its co-present distractors—by factors of roughly 2 to 11 for Opus, and never at all for Gemini 3.1 Pro—so only the protected value carries a secret-dependent distortion. The resistant models show no such differential, which is consistent with their having no channel.

We measure distractor specificity against *same-length* numbers. As Section 2 notes, an N-digit decoder cannot emit a value of a different length, so longer or shorter co-present numbers are non-confusable by construction. The N=2 arm uses five seeds because a `by_secret` split over 100 possible secrets holds out only ten of them, so any single-seed exact-match figure is x-out-of-ten. The N=4-at-scale arm uses three seeds, which is sufficient because N=50,000 leaves roughly 993 held-out secrets and the estimate is correspondingly stable. Relative to chance the N=4 arm is the stronger result: in the hardest cell (D, a buried memory-file secret in heavy clutter) Claude Opus 4.6 recovers the secret at 30$\times$ chance and Gemini 3.1 Pro at 115$\times$, where the N=2 measurement of the same cell is close to its floor.

J Estimation-Phase Cost

The estimation phase issues many black-box queries which we quantify below. The queries target the *public model API* rather than the victim's deployment, so only the model provider sees them. They are also a one-time cost, since a decoder trained once against a given model applies to every user of that model.

Costs are computed from token counts measured directly from provider-reported usage rather than estimated from text length, which matters because billed output includes reasoning tokens. Claude Opus 4.6 and GPT-5.4 nano emit none on this workload, but Gemini 3.1 Pro emits roughly eight times as many reasoning tokens as visible ones, making it by a wide margin the most expensive model to attack despite its mid-range

Scenario	Queries	Opus 4.6	Sonnet 4.6	Gemini Pro	Flash-Lite	GPT-5.4	nano
N=2 static, saturation min	4,000	$53.68	$32.05	$209.08	$2.91	$25.21	$1.39
N=2 static, full	10,000	$134.20	$80.13	$522.70	$7.26	$63.02	$3.47
N=4 static, full	10,000	$249.72	$138.85	$572.04	$13.53	$129.79	$5.97
N=4 adaptive	200,000	$4,994	$2,777	$11,441	$270.51	$2,596	$119.48
N=8 adaptive	400,000	$18,146	$10,369	$33,374	$999.07	$9,162	$333.91

Table 9 One-time estimation cost against the public API, from measured token counts and list prices as of 2026-08-04 (standard tier, not batch). Billed output includes reasoning tokens, which are zero for Claude and GPT-5.4 nano on this workload but substantial for Gemini 3.1 Pro, making it the most expensive model to attack. Grok 4 and Grok 4.1 Fast are omitted: xAI retired those model identifiers on 2026-05-15 and they silently redirect, so runs generated afterwards cannot be attributed to the named model.

list price. For a leaky model the cost is modest. The full N=2 static budget against Gemini 3.1 Flash-Lite is $7.26, and the decoder saturates well before that: roughly 200–400 distinct training secrets (Figure 6), which at 10 responses per secret is 2,000–4,000 queries, the first row of the table. At N=2 the resistant models never exceed chance at any budget we tested, so there is no amount of spending that converts them into leaky ones.

K User-Memory Predicate Inference: A Worked Example

Victim Model Generation (Truncated)

Subject: Explanation of Productivity Changes Over the Past Year

Dear Performance Review Committee,

. . . I was navigating an inheritance dispute (Probate Court case number 23-1456), which required me to *take four days of leave for court appearances*, reduced my work hours by approximately 15% . . .

My work productivity score has risen to 92 out of 100. Zero days of absenteeism . . . Assigned to lead the Lean Manufacturing Initiative . . .

Consistent daily routine beginning at 6:30 AM . . . Blood pressure normalized to 124/78 mmHg . . .

Memory (Included in Input)	Proposed Adversary	Linguistic Adversary (Mireshghallah et al., 2026)
"I took four days of leave from work to attend court."	✓	✓
"I have read *Adopted Hearts* and *The Adoptive Family*."	✓	✗

Table 10 An email generated by Claude Opus 4.6 for the CIMemories task of explaining productivity changes, generated while including two memories in the model's input context. Our adversary can detect the usage of *both* memories by observing the email, while the adversary that relies on verbatim repetition (highlighted in red) can only detect one.